

Project analysis report Part A

Background and Introduction

This project is an example of a design data factory floor with the first two workstations.

The aim is to extract and feature the data structure by filtering and calculating existing data (e.g., excel files).

Station #1: ETL. Extract data steam, Transform, and Load.

Station #2 Feature Engineering. Select relevant features. Reassemble and request new features. By selecting data from station #1, data visualization and deeper data analysis are achieved.

The code running platform is Spyder (Python 3.12), and the running input and output will be shown in the report.

Station #1 ETL

Description

This section processed four files of data.

- Using ASX200top10.xlsx, filtered out the daily price changes of PX_LAST (price) for ten stocks. After that, the average daily investment return of the investor over a five-year period from 15 to 20 years was calculated using this data frame.
- Using Client_Details.xlsx, I selected the group by age, risk, and investment ratio.
- Made the data frame of economics.
- Made the data frame of JSON

Data Input

```
ASX200top10_path = r'C:/Users/13769/FINS5545 data pool/ProjectA_Asset_Management/ASX200top10.xlsx'
df = pd.read_excel(ASX200top10_path, sheet_name=1, header=0, index_col=0, engine='openpyxl')
```

```
json_path = r'C:/Users/13769/FINS5545 data pool/ProjectA_Asset_Management/news_dump.json'
with open(json_path) as f:
    data = json.load(f)
    data = pd.DataFrame(data)
```

```
# Your given path
client_path = r'C:/Users/13769/FINS5545 data pool/ProjectA_Asset_Management/Client_Details.xlsx'
```

```
Economic_path = r'C:/Users/13769/FINS5545 data pool/ProjectA_Asset_Management/Economic_Indicators.xlsx'
df_E = pd.read_excel(Economic_path, sheet_name=0, header=0, index_col=0, engine='openpyxl')
```

Data Filtering and Calculations

For the ASX200top20

Screened daily prices and average daily stock price returns for ten stocks over 15 to 20 years (with 2015-03-30 nulls removed in the return calculation)

批注 [渭何1]: 抓取、转化、提炼数据。加载不同形式的数据流，通过网站提供的 API 得到数据，处理成为后续想要的形式。

Calculate the random weighted average return on investing in those ten stocks.

For client detail

Filter records with a screening risk level of 4 and above

Filter record with a screening age group of less than 4

Filter records with a CSL equity asset distribution of more than 50%

Data Output

After filtering, get data frame about ten stocks and the expected average return information of 10 stocks.

```
Dates      BHP AT Equity  CSL AT Equity  ...  BXB AT Equity  FPH AT Equity
2015-03-30    28.1285      91.60      ...      11.50      6.40
2015-03-31    28.9976      92.12      ...      11.52      6.44
2015-04-01    28.1528      92.63      ...      11.48      6.38
2015-04-02    28.2406      93.32      ...      11.54      6.43
2015-04-03    28.2400      93.32      ...      11.54      6.43
...          ...          ...          ...          ...
2020-03-27    29.0300      279.12     ...      18.31      27.27
2020-03-30    30.2000      312.88     ...      18.61      30.23
2020-03-31    28.9800      296.68     ...      19.56      28.84
2020-04-01    30.2300      307.00     ...      11.19      29.78
2020-04-02    29.8500      308.27     ...      11.57      29.60
[1309 rows x 10 columns]
```

```
Dates
2015-03-31    0.015176
2015-04-01   -0.012005
2015-04-02   -0.004356
2015-04-03    0.000000
2015-04-06    0.000000
...
2020-03-27   -0.041061
2020-03-30    0.050070
2020-03-31   -0.045149
2020-04-01    0.038175
2020-04-02   -0.010553
Length: 1308, dtype: float64
```

Also, the following data frames were built.

A data frame on client detail

A data frame on JSON

A data frame on economic

Definition of station#1

ETL stands for Extract, Transform, Load. It's a data processing method that is extracted from various sources, transformed into a standardized format, and loaded into a target system like a database or data warehouse for analysis and reporting.

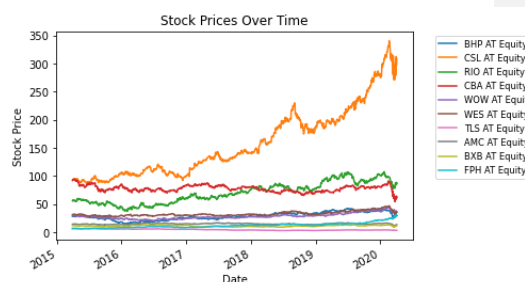
Station #2 Feature Engineering

Aim at grasping some data features and representing the data features through data visualization.

Stock Trend Features

The most direct way to make a stock price chart is to make a data frame of 10 stocks based on station#1.

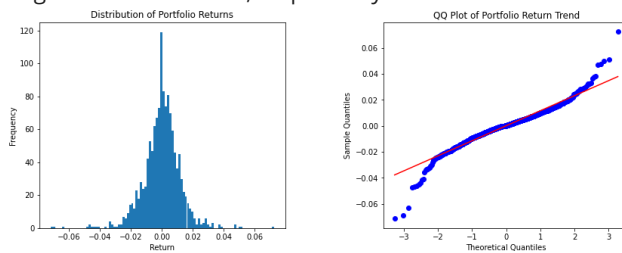
It can be observed that the average share price has been trending upward for the last five years, with CSL's share price rising the most. Looking at the volatility of the graph, those with higher volatility are reckoned to have risen more. This aligns with the investment philosophy that the higher the risk, the higher the return.



批注 [渭何2]: 数据挖掘、数据可视化。1. 将资产的特征提取出来, 利用统计学的方法可视化数据特征。2. 根据客户现有资产头寸, 结合聚类分析算法, 自动判断客户风险偏好类型。

Return distribution

Using the return trend data obtained from station#1, the distribution and QQ plots are designed for the returns, respectively.



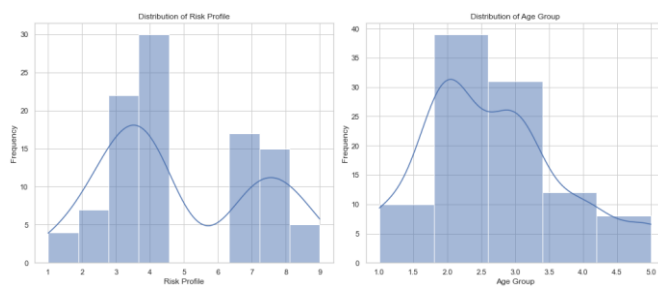
Also, to test further whether the returns match a normal distribution.

```
Skew of data set      -0.333
Skew test p-value      0.000
Kurt of data set       4.822
Kurt test p-value      0.000
Norm test p-value      0.000
```

This result indicates that the return on investment satisfies a normal distribution. (P-Value<0.05)

Distribution of Age and Risk Appetite

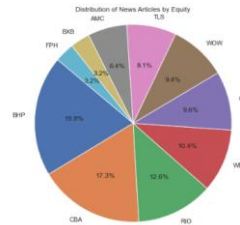
The investment distribution and age distribution of the clients are made based on the data from client details,



Frequency of News

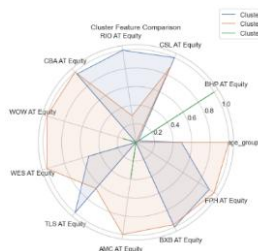
Analyze the frequency of news appearances of companies based on JSON files.

As can be seen, the companies with larger market capitalization among them are discussed more.



K-means clustering analysis

Through the application of K-means clustering analysis, the customer base is categorized into three distinct clusters, as follows:



Cluster 0 contains 52 customers,
Cluster 1 contains 25 customers,
Cluster 2 contains 23 customers.

Cluster 0: The client has the highest average age (about 3.29), and the investment portfolio is balanced among stocks. This cluster may represent a group of senior or aged customers with more diversified investment preferences and looking for a stable investment.

Cluster 1: This cluster has the lower average age group (about 2.48) and particularly has a significantly higher ratio of investments in RIO AT Equity than other equities; this group may represent a group of senior clients who are seeking higher returns in specific industries or companies.

Cluster 2: This cluster has the lowest average age group (about 1.57) and has a very high percentage of investments in BHP AT Equity and a low percentage of investments in other equities, especially BXB AT Equity and FPH AT Equity, which have almost no investments. This may indicate a younger client group with highly concentrated investments in a single stock or sector.

Definition of station#2

Feature engineering uses domain knowledge to create new features from raw data that make machine learning algorithms work more effectively. It involves transforming raw data into meaningful variables that help improve predictive models' performance. This can include creating new data columns, modifying existing ones, combining data from different sources, or encoding categorical variables. The goal is to provide solid and relevant inputs for machine learning models to enhance their accuracy and performance.

Project analysis report Part B

Introduction

This project is an example of a design data factory floor with the latter two parts. Obviously, this is the more challenging part.

The aim is to allocate the portfolio through customers' risk appetite, and adjust the portfolio allocation under economic conditions and market information changes. Finally, modelling the market movements (indices) to make predictions. Station #3 Moedl Design. Building models that make customer allocate their portfolio.

Station #4: Model Implementation Implement the model and develop a flow of customer service workflows.

The code running platform is Spyder (Python 3.12), and the running input successfully and output will be shown in the report.

Station #3: Model Design

Description

Several models have been made in this section to help clients optimise their asset allocation.

Use Monte Carlo simulation to produce a risk-return distribution with random weights.

Make an efficient frontier that maximises returns with the same risk. (Optimizing Monte Carlo simulations)

Match risk preferences to portfolios

Combine market information and economic indicators for optimal portfolio adjustment

Market trend (index) forecasting with ARIMA

model designed and aim

1. Monte Carlo simulation

Generate a portfolio of assets under different weights in random weights. The aim is to explore the boundaries of risk and reward initially. This simulation shows the limitations of ordinary customers without an optimised algorithm. (Harrison, 2010)

2. Portfolio Optimization Models

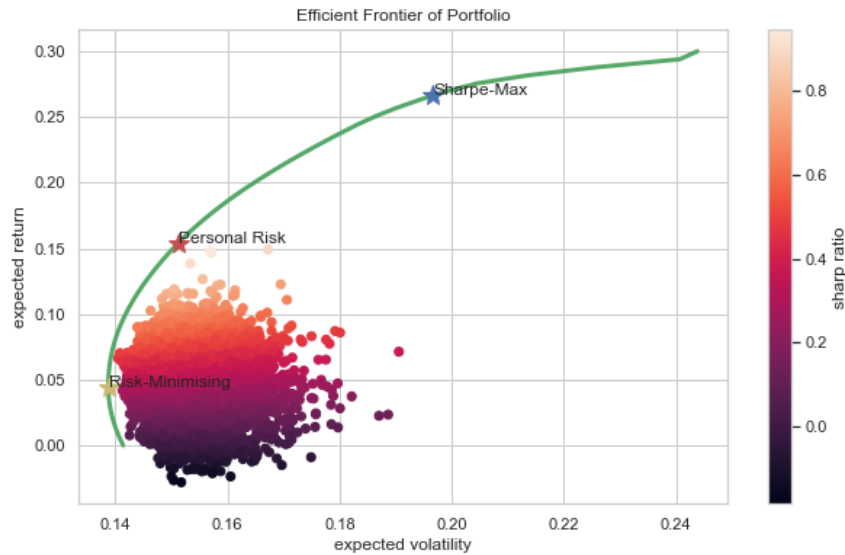
Making an efficient frontier aims to show the maximum return and minimum risk that can be achieved under the optimal algorithm. (Bailey & , 2012)

批注 [渭何3]: 1. 蒙特卡洛模拟

2. 最优化资产组合模型

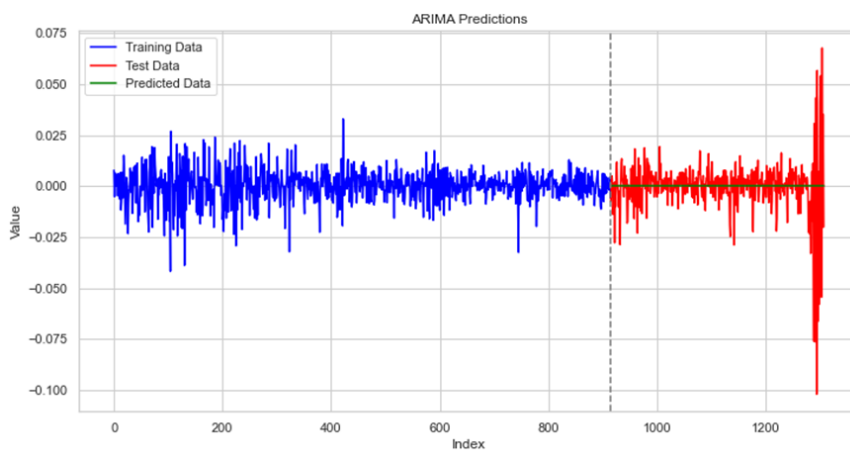
3. ARIMA 判断股指

4. 基于 NLP 模型的市场情感分析



3. ARIMA

Forecasting market trends (indices) with ARIMA

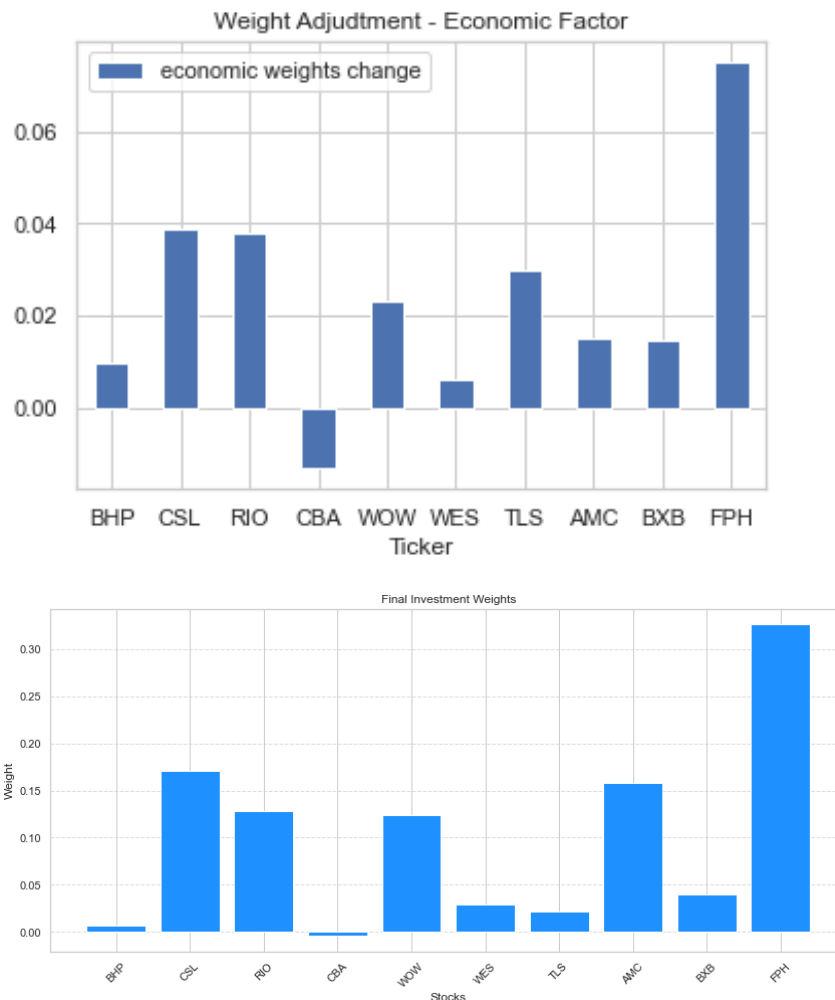


4. NLP (combine with OLS model)

OLS forecasts the impact of economic indicators on markets. Estimate the change in current stock price by economic factors.

5. Sentiment Analytics

Sentiment analysis and categorisation of published messages. Positive news proves to be beneficial to the company's share price, while negative news can cause the company's share price to fall. (Chowdhury, et al., 2014)



model assumptions

Market Efficiency Assumption:

This assumption is closely related to the Efficient Frontier and Monte Carlo Simulations. The market efficiency assumption posits that all available information is already reflected in asset prices, which is fundamental for making investment decisions using these methods.

Risk Preferences and Constraint Satisfaction:

The model assumes that investors' risk preferences can be accurately quantified and mapped to specific portfolio strategies. This is closely related to optimising the portfolio to maximise the Sharpe ratio or minimise variance.

Accuracy of Sentiment Analysis:

It is assumed that the sentiment scores derived from financial news are accurate indicators of market sentiment and that these sentiments have a predictable impact on stock prices. This is crucial for adjusting portfolio weights based on sentiment analysis.

computing speed and accuracy issues

This model is only applicable to 10 stocks; including more stocks in the future could lead to slower processing times, especially with large datasets or an increased number of simulations, such as those in Monte Carlo simulations.

Accuracy and Sensitivity:

Since the model involves many parameters, its accuracy is influenced by assumptions of market efficiency and the normal distribution of returns. Any deviation in actual data from these assumptions can lead to inaccurate forecasts and suboptimal portfolio allocations. Moreover, the model's sensitivity to the accuracy of input data, such as financial news sentiment or economic indicators, adds complexity and potential for error.

The expectation of model implementation

- Risk Category Definition

High-Risk Group: Clients willing to accept high volatility in exchange for potentially higher returns.

Risk-Neutral Group: Clients who prefer a balance between risk and return.

Low-Risk Group: Clients prefer stability and lower volatility, even at the cost of lower returns.

- Optimal Risk-Return Calculation

Develop a portfolio for each risk category using historical data to maximise the expected return for a given level of risk. This might involve using the Sharpe Ratio for the high-risk group to maximise returns relative to volatility and minimizing volatility for the low-risk group.

- Dynamic Adjustment Based on Client Preferences

Adjust the portfolios according to individual client risk preferences within each group. For example, if a client in the risk-neutral group prefers slightly higher risk, you could adjust their portfolio to include higher-risk assets.

Example: Risk-Neutral Group Adjustments

Adjusting Investment Weights Based on Market Sentiment: Use sentiment analysis from financial news to gauge market mood and adjust the weights of investments in the risk-neutral portfolio accordingly. For instance, increase the weight in stocks if the sentiment is positive.

- **Adjusting Investments Based on Economic Indicators:**

Incorporate economic data such as GDP growth rates, unemployment figures, or consumer confidence indices further to fine-tune the investment allocations in the risk-neutral portfolio.

- **Market Index Trend Forecasting**

Utilise ARIMA (AutoRegressive Integrated Moving Average) models to forecast future trends of relevant market indices. This forecast can help in setting expectations for portfolio performance and making informed adjustments.

- **Visualization and Output**

Create visualisations such as line graphs for ARIMA predictions, bar charts for portfolio allocations, and scatter plots for risk vs. return profiles.

Boundaries of the model

Constraints

In this context, the sum of the elements in the array of asset weights must equal 1. This constraint ensures that all asset weights add up to 100%, indicating that all available investment funds are fully allocated across various assets without any remainder. This constraint is a fundamental condition in portfolio management to ensure the integrity and practicality of the portfolio.

Bounds

This line of code sets bounds for every variable (asset weights) in the optimisation problem. `bnds` is a tuple generated through list comprehension, defining a closed interval from 0 to 1 for each asset weight. This means that each asset's weight cannot be less than 0% (to avoid short selling) and cannot exceed 100% (to prevent over-concentration of funds in a single asset). These boundary settings ensure that all weights are within a reasonable and executable range.

Optimal Risk Portfolio Curve

This curve limits the boundaries of risk and return; to achieve higher returns, one would need to engage in risk-free borrowing. In practice, one cannot borrow infinitely at the same interest rate; hence, this discussion temporarily disregards scenarios involving loan investments or partial investments.

These settings aim to ensure that all solutions in portfolio optimisation are rational, conform to real investment operations, and can be effectively searched for optimal solutions through algorithms. Constraints and bounds are key to achieving this objective, helping the optimisation algorithm focus within a feasible and reasonable space.

Define Station #3

Design several different models to predict changes in stock prices. Receive data from station #3 and do in-depth feature engineering. Make adjustments to asset

allocation in response to market economic indicators, user risk appetite, and market news.

Station #4: Model Implementation

Product Design

First, the data are analysed according to the model of station#3. Design three types of asset allocation with different risk preferences: low-risk portfolio, medium-risk portfolio, and high-risk portfolio. Then, produce a set of workflows that will serve the clients.

Steps when implementing this solution

1. Design product structure

Market research: Conduct thorough market research to understand the current market trends, potential competitors, and customer demands. This research should focus on identifying the popular asset classes and investment strategies. Set the risk level: Design the product to cater to three different risk levels (high, medium, and low), allowing for personalised portfolio management based on risk tolerance. Develop strategies and asset allocations corresponding to these risk levels, ensuring they are diversified and aligned with typical risk profiles.

2. Advertise

Develop a marketing campaign highlighting your product's unique features, such as personalised risk management and ease of use. Utilise digital marketing, social media, and possibly traditional advertising methods to reach a broad audience.

Target: Target ads to specific demographic groups likely interested in investment solutions, such as working professionals, retirees, or savvy investors looking for innovative tools.

3. Questionnaires and interviews with clients

Questionnaires : Create detailed surveys to collect data on potential customers' investment experience, risk tolerance, financial goals, and preferences. Use this data to refine your product features and user experience.

Talk to clients: Conduct in-depth interviews with a select group of potential users to gain qualitative insights into their needs and expectations from an investment tool. This can help in adjusting the product to fit the market needs better.

4. giving the advice

advice: Based on the collected data from surveys and interviews, provide personalised investment recommendations that align with the user's risk profile and financial goals.

5. choose stock: Allow customers to choose from a curated list of stocks, ensuring these options are diversified across different sectors and risk

批注 [渭何4]: 产品落地实施: 1. 提供不同的产品给不同风险偏好的客户。设计产品结构、销售渠道、广告、客户群体分析、提供投资咨询建议。2. figma 体现整体的产品结构, 吸引客户。

categories. This choice empowers users and enhances their engagement with the tool.

6. allocate the asset under model

Allocate the asset: Implement a sophisticated asset allocation model that uses algorithms to distribute investments across the chosen stocks based on the customer's finalised risk preference. This model should optimise for expected returns, risk, and customer investment strategy compliance.

6. giving the investment adverse (final weight assets)

Final weight: After the asset allocation model has run, present the final investment weights in each selected stock. These weights show how much of the client's capital is invested in each asset, tailored to maximize returns while managing risk.

Output the results: Use visualizations such as bar charts or pie charts to make the results understandable and visually appealing. This helps customers easily grasp the distribution of their investments.

Difficulties

When I tried to predict the overall index changes using ARIMA, I found that the index (or stock) prices are not stable time series data. Therefore, effective predictions cannot be made. Moreover, ARIMA is only suitable for single time series data and cannot be used to predict index changes by combining other stock price changes.

Unfortunately, when I tried to use other models, they often failed to run successfully, and the results did not meet my expectations. I only made predictions about index changes using ARIMA.

When I explain why, it involves explaining a lot of professional financial knowledge, such as optimal risk configuration, efficient frontier, etc. My model is based on these designs. When I explain to clients why I configured the assets this way, they might not understand

Steps introducing the product to your clients

1. Introduce the Efficient Frontier: Explain the concept of achieving the best possible returns for a given level of risk using Modern Portfolio Theory.
2. Explain the Sharpe Ratio: Describe how this ratio measures the return of an investment relative to its risk, with a focus on maximizing returns per unit of risk.
3. Assess Client Risk Profile: Evaluate the client's financial goals, risk tolerance, and investment timeline to tailor investment strategies.
4. Align Asset Allocation: Match the client's risk tolerance with appropriate asset distribution to optimise the balance between risk and return.

5. Optimise Portfolio: Regularly adjust the portfolio to enhance performance, manage risk, and adapt to changing market conditions.
6. Provide Strategic Advice: Offer actionable investment recommendations based on the client's objectives and market opportunities.

Customer journey

1. Discovery and Engagement: Clients discover the service through online ads or sales outreach and are drawn by its unique benefits, such as personalized investment strategies.
2. Onboarding and Risk Profiling: Clients fill out a detailed questionnaire or engage in interviews to identify their financial goals and risk preferences, which are essential for tailoring the service.
3. Stock Selection: Clients select ten stocks they are interested in, with support and educational materials provided to help them make informed decisions.
4. Asset Allocation Using Models: Investments are allocated using a sophisticated model based on the client's chosen stocks and risk profile, with detailed explanations provided to ensure transparency.
5. Monitoring and Adjustments: The portfolio is continuously monitored and adjusted in response to market changes or shifts in the client's financial goals, with ongoing communication to keep clients informed.
6. Feedback and Iterative Improvement: Regular feedback is collected to enhance the service, with adjustments made to meet evolving client needs and market conditions.
7. Long-term Value and Relationship Building: The focus is on delivering sustained value and building a long-term relationship through consistent performance and client satisfaction,

Wireframes / FIGMA design

A flow chart of the data processing and results was designed. Details are in Figma file

Define Station #4

This part requires outputting our results to the client. I think this may be more difficult than designing models because we need to simplify some complex financial theories.

In addition, designing a workflow to serve clients. Because the model itself does not attract clients.

reference

Bailey, D. H. & M. M., 2012. THE SHARPE RATIO EFFICIENT FRONTIER. *Journal of risk*.

Chowdhury, S. G., Routh, S. & Satyajit, C., 2014. News Analytics and Sentiment Analysis to Predict Stock Price Trends. *International Journal of Computer Science and Information Technologies*.

Harrison, 2010. Introduction To Monte Carlo Simulation.